

Caleb Kang

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science in Computer Science, Grainger College of Engineering

Aug. 2026 – May 2028

- **In Progress (Fall 2026):** Computer Architecture (CS 233), Database Systems (CS 411), Software Design Lab (CS 222), Probability & Statistics for Computer Science (CS 361)

William Rainey Harper College

Palatine, IL

Associate in Engineering Science, GPA: 3.88/4.0

Aug. 2024 – May 2026

- **Completed Coursework:** Data Structures & Algorithms, Linear Algebra, Differential Equations, Discrete Mathematics
- **Awards:** Motorola Solutions Foundation Scholarship (Full-Ride), Engineering Pathways Scholarship, President's List

EXPERIENCE

Software Development Engineer Intern

May 2026 – Aug. 2026

Amazon

Bellevue, WA

- Built an LLM investigation agent for Teller, Amazon's Accounts Payable system (\$2.5B+ processed), turning analyst questions into validated SQL over 130M+ production records
- Hardened the tool layer beneath the LLM with a read-only SQL validator, table allowlist, Bedrock guardrails, and anti-hallucination grounding checks; shipped 100+ unit and integration tests
- Automated anomaly detection with z-score baselines and a 20+ tool-call investigation loop; surfaced a production void batch of 63,469 payments (z-score 62) in 2–3 minutes
- Deployed the containerized agent on Bedrock AgentCore with AWS CDK (TypeScript) infrastructure as code, CI/CD, and least-privilege cross-account IAM access to SOX-regulated data

Software Engineer Intern

Jun. 2025 – Aug. 2025

Fermi National Accelerator Laboratory

Batavia, IL

- Built a fault-tolerant Python state machine to orchestrate ASIC test workflows for the Deep Underground Neutrino Experiment (DUNE), supporting validation of approximately 300,000 chips across six national labs
- Mapped state transitions and failure paths in UML, implemented error handling, and defended the architecture in peer reviews and technical presentations to research scientists

Student Researcher

Oct. 2024 – May 2025

Biola University (Remote)

La Mirada, CA

- Fine-tuned and benchmarked a multimodal authentication pipeline (CLIP, Whisper, Sortformer) in PyTorch on NVIDIA NeMo
- Built evaluation workflows measuring inference latency, throughput, and edge-case reliability across vision, audio, and speaker-diarization inputs

PROJECTS

C++ Farming Simulator | C++, OOP, Memory Management, Catch2

Oct. 2025 – Dec. 2025

- Built a terminal-based simulation on a dynamic 2D pointer grid, handling heap allocation and deallocation manually across gameplay state changes
- Covered state invariants and edge cases in movement and resource logic with a deterministic Catch2 test suite

Duckiebot Robotics Initiative | Linux, Docker, C++, Raspberry Pi, NVIDIA Jetson

Dec. 2024 – Jan. 2025

- Led a 4-person team building autonomous rover controls in C++; owned sprint planning and deployment validation checks
- Deployed the Duckietown stack in Docker on embedded Linux (Raspberry Pi, NVIDIA Jetson), debugging kernel-level networking, serial communication, and sockets

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, TypeScript, JavaScript

Systems & Infrastructure: Linux/Unix, Docker, Git, CI/CD, AWS (Athena, Glue, Lake Formation, S3, Bedrock, CDK)

Data & ML: Relational Databases, PyTorch, NVIDIA NeMo, Strands Agents SDK, Claude Code

Concepts: Data Structures & Algorithms, OOP, State Machines, Memory Management, Containerization, Unit Testing